

Predicate Logic.

The number $x+2$ is an even integer $\therefore p(x)$

↳ **Open statements**

- they contain one or more variables.
- they themselves are not propositional statements
- They become propositional statements when the variables assume some values.

→ $p(\underline{x})$ / $q(\underline{x})$. . . / $r(\underline{x}, \underline{y})$
Variables.

$q(x, y)$: The numbers $y+2$, $x-y$ & $x+2y$ are even integers.

$p(5)$: $x+2 = 5+2 = 7 \neq \text{even}$. $\rightarrow \text{False}$

$p(4, 2)$: True

$\neg p(5)$: True.

$p(x)$: $2x$ is an even integer.
true for all values of x .

Some values of x] $\rightarrow$ quantifiers.
vs.

all values of x]

Some $\rightarrow$ existential quantifiers
"there exists some x "

$$\exists x p(x)$$

bounded statement.

all $\rightarrow$ universal quantifiers
"for all"

$$\forall x p(x)$$

$\left\{ \begin{array}{l} p(x) \\ \text{sometimes true} \\ \text{sometimes false} \end{array} \right\} \quad \neg p(x) \quad \left\{ \begin{array}{l} \text{sometimes true} \\ \text{sometimes false} \end{array} \right\}$

$$\exists x p(x)$$

always

$$\exists \neg p(x)$$

(always)

well defined.

$q(x, y)$ — for some x & some y .

either $\exists x, \exists y q(x, y)$ } same
 or $\exists x, y q(x, y)$

$\forall x, \forall y q(x, y)$ } same.
 or $\forall x, y q(x, y)$

Recap:
 in $q(x)$ $\rightarrow$ x is a free variable
 in $\exists x q(x)$ or $\forall x q(x)$ $\rightarrow$ x is a bounded variable.

Ex. 1:

$$\begin{aligned} p(x) &: x \geq 0 \\ q(x) &: x^2 \geq 0 \\ r(x) &: x^2 - 3x - 4 \geq 0 \\ s(x) &: x^2 - 3 > 0 \end{aligned}$$

Truth value:

$$\forall x [p(x) \rightarrow q(x)]$$

$\downarrow$
True.

$$\exists x [p(x) \wedge r(x)]$$

$\downarrow$
True.

$$\forall x [q(x) \rightarrow s(x)] \rightarrow \text{false} \rightarrow \text{counter example. } (x=1).$$

$$\forall x p(x) \Rightarrow \exists x p(x)$$

logically implies.

↓

difference of this form $\xrightarrow{\text{implies.}}$

$\exists x p(x)$ is true whenever $\forall x p(x)$ is true.

E.g. 2.

$p(x)$: x is a rational number.

$q(x)$: x is a real number.

$$\forall x [p(x) \rightarrow q(x)].$$

$$\forall x [q(x) \rightarrow p(x)] \quad \times$$

E.g. 3

$e(t)$: Triangle is equilateral

$a(t)$: Triangle t has all three angles $= 60^\circ$.

$$\forall t [e(t) \leftrightarrow a(t)]. \quad \text{biconditional}$$

Statement	When true?	When false?
$\exists p(x)$	For some 'a' (at least one) $p(a)$ is true.	For every every 'a' in the universe, $p(a)$ is false.
$\forall x p(x)$	For every 'a', $p(a)$ is true.	For at least one 'a' in the universe, $p(a)$ is false.
$\exists x \neg p(x)$	For at least one choice of 'a', $p(a)$ is false or in other words $\neg p(a)$ is true.	For all 'a' in the universe $p(a)$ is true.
$\forall x \neg p(x)$	For every 'a' in the universe $p(a)$ is false.	For at least one 'a' in the universe where $\neg p(a)$ is false or $p(a)$ is true.

Logical equivalence.

$\forall x [p(x) \Leftrightarrow q(x)] \rightarrow$ logical equivalence.
 $\hookrightarrow p(x) \leftrightarrow q(x)$ is true for every $x=a$ in the universe.

$\forall x [p(x) \Rightarrow q(x)] \rightarrow$ logical implication.
 $p(x) \rightarrow q(x)$ is true for every $x=a$ in the universe.

$\forall x [p(x) \Leftrightarrow q(x)]$ iff $\forall x [p(x) \Rightarrow q(x)]$ AND $\forall x [q(x) \Rightarrow p(x)]$.

E.g. 4.

for $n = 1$ to 20 do

$$A[n] = n * n - n$$

① $\forall n [A[n] \geq 0] \rightarrow \text{true.}$

② $\forall n [1 \leq n \leq 19] \rightarrow (A[n] < A[20])].$
 $\hookrightarrow \text{true.}$

③ $\exists n (A[n+1] = 2A[n]).$ $n = 3.$

④ $\forall m, \forall n [(m \neq n) \rightarrow (A[m] \neq A[n])] \rightarrow \underline{\text{true.}}$

⑤ $\forall m, n [(m < n) \rightarrow (A[m] \neq A[n])] \rightarrow \text{true.}$

For $p(x)$ & $q(x)$, and $\rightarrow$

① Contraposition:

$$\forall x [p(x) \rightarrow q(x)] \checkmark$$

contraposition $\hookrightarrow \forall x [\neg q(x) \rightarrow \neg p(x)]$.

② Converse:

$$\forall x [p(x) \rightarrow q(x)]$$

converse $\hookrightarrow \forall x [q(x) \rightarrow p(x)]$.

③ Inverse:

$$\forall x [p(x) \rightarrow q(x)]$$

inverse $\hookrightarrow \forall x [\neg p(x) \rightarrow \neg q(x)]$.

H.W.

$S(x)$: x is a square
 $e(x)$: x is equilateral.

Assent:

(a) $\forall x [S(x) \rightarrow e(x)]$
 (b) $\forall x [e(x) \rightarrow S(x)]$.

(c) $\forall x [\neg S(x) \rightarrow \neg e(x)]$.

(d)

Any connection between (b) & (c)?